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Inventor(s)

Shulepov; Sergei Yuryevich et al.

DETERMINING A FLOW PROFILE IN AN ARTERY BASED ON ULTRASOUND IMAGING DATA

Abstract

The invention relates to determining a flow profile in an artery from ultrasound imaging data. For this end, a computer-implemented method (300) is provided for determining (330) a flow profile based on an artery dimension, a velocity in the artery and a measure of asymmetry, wherein the artery dimension, the velocity in the artery and the measure of asymmetry are determined (320) from ultrasound imaging data comprising B-mode, pulsed wave Doppler, and color or power Doppler ultrasound imaging data.

Inventors: Shulepov; Sergei Yuryevich (Eindhoven, NL), Joshi; Rohan (Eindhoven, NL), van Rooij; Johannes Antonius (Best, NL)

Applicant: KONINKLIJKE PHILIPS N.V. (EINDHOVEN, NL)

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Background/Summary

FIELD OF THE INVENTION

[0001] The invention relates to determining a flow profile in an artery based on ultrasound imaging data.

BACKGROUND OF THE INVENTION

[0002] When determining blood flow in an artery using ultrasound, a sonographer typically positions an ultrasound probe with the imaging plane in the middle of the artery and along the main axis of the flow. During the examination, the sonographer then typically utilizes two types of measurements: a color or power Doppler measurements to assess the flow distribution over the lumen of the artery, and a pulsed wave Doppler measurement to obtain the blood velocity waveform. An example method for continuous non-invasive monitoring of multiple arterial parameters of a patient is provided in US 2012/0078106. This procedure, however, assumes that the velocity distribution is axially symmetric, and the maximum velocity is close to the artery or lumen center. However, most of the arteries are not straight, and tortuosity of the arteries also increases with age of the patient.

SUMMARY OF THE INVENTION

[0003] It is, inter alia, an object of the invention to determine a flow profile in an artery with increased accuracy, and in particular in arteries with increased tortuosity.

[0004] The invention is defined by the independent claims. Advantageous embodiments are defined in the dependent claims.

[0005] According to an aspect of the invention, there is provided a computer-implemented method for determining a flow profile in an artery based on ultrasound imaging data, the method comprising: [0006] receiving ultrasound imaging data of an artery in triplex mode, wherein the triplex mode comprises: [0007] B-mode ultrasound data, [0008] pulsed wave Doppler ultrasound data, and [0009] color or power Doppler ultrasound data; [0010] determining (320), from the ultrasound imaging data, [0011] an artery dimension; [0012] a velocity in the artery cross-section, and [0013] a measure of asymmetry of the flow profile; and [0014] determining a flow profile from [0015] the artery dimension, [0016] the velocity in the artery cross-section, and [0017] the measure of asymmetry of the flow profile.

[0018] In this manner, by obtaining a few measurement points from ultrasound imaging data (i.e., an artery dimension, a velocity in the artery cross-section, and a measure of asymmetry of the flow), an accurate flow profile can be determined according to analytical expressions. In other words, the flow profile is re-constructed with the knowledge of parabolic flows and the parameters determined from the ultrasound imaging data. The method may further be carried out during or after an ultrasound imaging procedure.

[0019] The method of may further comprise: [0020] determining a local artery curvature based on [0021] the artery dimension, [0022] the flow profile, and [0023] the measure of asymmetry of the flow profile.

[0024] By determining a local artery curvature, it is possible to improve for example methods for plaque removal and/or stent placements.

[0025] In an example the artery dimension is determined based on the B-mode ultrasound data. In an example the velocity in the artery cross-section is determined based on the pulsed wave Doppler ultrasound data. In an example, the velocity in the artery cross-section is determined in a center of

the artery cross-section. In an example, the measure of asymmetry of the flow profile is determined based on the color or power Doppler ultrasound data.

[0026] In an example, the measure of asymmetry comprises determining a location of the maximum flow velocity in an artery cross-section; and based on said location, determining a parameter representative of a deviation of said location with respect to the center of the artery cross-section.

[0027] In an example, the parameter representative of the deviation of the location of the maximum flow velocity in an artery cross-section is the Dean number.

[0028] According to an aspect of the invention, there is provided a computer program product comprising instructions for enabling a processor to carry out the above method. The computer program product may be software available for download from a server, e.g., via the internet. Alternatively, the computer program product may be a suitable (non-transitory) computer readable medium on which the instructions are stored, such as an optical storage medium or a solid-state medium supplied together with or as part of other hardware.

[0029] According to an aspect of the invention, there is provided a system for determining a flow profile from ultrasound imaging data, the system comprising a processor configured to carry out the above method.

[0030] In an example, the system may further comprise [0031] an array of acoustic elements in communication with the processor, and [0032] wherein the processor is configured to control the array of acoustic elements to send and receive acoustic signals to generate ultrasound imaging data.

[0033] In an example, the system is a hemodynamic monitoring patch.

[0034] These and other aspects of the invention will be apparent from and elucidated with reference to the embodiments described hereinafter.

Definitions

[0035] “cross-section” refers to an intersection of an artery in three-dimensional space with an imaging plane. A “cross-section” of the artery is thus any plane intersecting the artery itself, not necessarily at a 90-degree angle.

[0036] “Transversal cross-section” is used to refer to a plane intersecting the artery, wherein said plane forms a 90 degree angle with the length of the artery. A cross-section of the artery thus includes the possibility of a transversal cross-section.

[0037] “maximum velocity” is used to refer to the maximum velocity encountered in a velocity profile in a transversal cross-section of an artery.

[0038] “peak velocity” is used to refer to the peak velocity in a transient (i.e., evolving in time) velocity profile at a location. For example, a peak in a transient velocity profile from pulsed wave Doppler data.

[0039] “center-line” refers to the symmetry axis along the length of the artery. In other words, the axis of the cylinder that represents an artery.

[0040] “shift in maximum velocity” or “maximum velocity shift” or “velocity maximum shift” or any other combination of these words, refers to the shift of the maximum velocity in a velocity profile with respect to the centroid of the artery cross-section. In other words, the distance between the location of the velocity maximum and the central axes of the artery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] FIG. 1 shows a schematic of an ultrasound system in which the invention may be used.

[0042] FIG. 2 shows a schematic of a processing circuit that may be used in an embodiment.

[0043] FIG. 3 shows a diagram of a method according to an embodiment.

[0044] FIG. 4 shows an exemplary B-mode ultrasound image.

[0045] FIG. 5 shows an exemplary pulsed wave Doppler ultrasound image.

[0046] FIG. 6A shows an exemplary flow contour in an artery.

[0047] FIG. 6B shows an exemplary flow profile in an artery.

[0048] FIG. 7 shows an exemplary arterial branch.

[0049] FIG. 8 shows an exemplary color Doppler ultrasound image.

[0050] FIG. 9 shows an exemplary relationship between the maximum velocity shift and the Dean number.

[0051] FIG. 10 shows an exemplary flow profile.

DESCRIPTION OF EMBODIMENTS

[0052] The invention will be described herein with reference to the figures.

[0053] It should be understood that the following description, while indicating exemplary embodiments, is intended for purposes of illustration only, and not intended to limit the scope of the invention. It should also be understood that the Figs are mere schematics and not drawn to scale. It should also be understood that the same reference numerals are used throughout the Figs to indicate the same or similar parts.

[0054] The invention provides a computer-implemented method and system to carry out said computer-implemented method, to determine a flow profile within an artery from ultrasound imaging data. The flow profile may be a 2D flow profile and/or a 3D flow profile. According to an exemplary embodiment, the computer-implemented method may also be used to determine a curvature of an artery. As disclosed herein, the method may be carried out by a system during image acquisition, i.e., in real time, or may be carried out a posteriori in a subsequent step, at any time after image acquisition.

[0055] FIG. 1 shows a schematic diagram of an exemplary ultrasound imaging system **100**. The system **100** includes an ultrasound imaging probe **110** in communication with a host **130** over a communication interface or link **120**. The probe **110** may include a transducer array **112**, a beamformer **114**, a processor **116**, and a communication interface **118**. The host **130** may include a display **131**, a processor **136**, a communication interface **138**, and a memory **133**. The host **130** and/or the processor **136** of the host **130** may also be in communication with other types of systems or devices in replacement or addition to the here mentioned systems such as for example, an external memory, external display, a subject tracking system, an inertial measurement unit, etc. It is understood that the beamformer may also be a microbeamformer. It is further understood that the components as shown here may also be configured in alternate arrangements. For example, the processor **116** and/or the beamformer **114** may be located outside of the probe **110** and/or the display **131** and/or memory **133** may be located outside of the host **130**.

[0056] In some embodiments, the probe **110** is an external ultrasound imaging device including a housing configured for handheld operation by a user. The transducer array **112** can be configured to obtain ultrasound data while the user grasps the housing of the probe **110** such that the transducer array **112** is positioned adjacent to or in contact with a patient's skin. The probe **110** is configured to obtain ultrasound data of anatomy within the patient's body while the probe **110** is positioned outside of the patient's body. In some embodiments, the probe **110** can be a patch-based external ultrasound probe. For example, the probe may be a hemodynamic patch.

[0057] In other embodiments, the probe **110** can be an internal ultrasound imaging device and may comprise a housing configured to be positioned within a patient's body, including the patient's coronary vasculature, peripheral vasculature, esophagus, heart chamber, or other body or body cavity. In some embodiments, the probe **110** may be an intravascular ultrasound (IVUS) imaging catheter or an intracardiac echocardiography (ICE) catheter. In other embodiments, probe **110** may be a transesophageal echocardiography (TEE) probe. Probe **110** may be of any suitable form for any suitable ultrasound imaging application including both external and internal ultrasound imaging.

[0058] The transducer array **112** emits ultrasound signals towards an anatomical object **105** of a

patient and receives echo signals reflected from the object **105** back to the transducer array **112**. The transducer array **112** can include any suitable number of acoustic elements, including one or more acoustic elements and/or a plurality of acoustic elements. In some instances, the transducer array **112** includes a single acoustic element. In some instances, the transducer array **112** may include an array of acoustic elements with any number of acoustic elements in any suitable configuration. For example, the transducer array **112** can include between 1 acoustic element and 10000 acoustic elements, including values such as 2 acoustic elements, 4 acoustic elements, 36 acoustic elements, 64 acoustic elements, 128 acoustic elements, 500 acoustic elements, 812 acoustic elements, 1000 acoustic elements, 3000 acoustic elements, 8000 acoustic elements, and/or other values both larger and smaller. In some instances, the transducer array **112** may include an array of acoustic elements with any number of acoustic elements in any suitable configuration, such as a linear array, a planar array, a curved array, a curvilinear array, a circumferential array, an annular array, a phased array, a matrix array, a one-dimensional (1D) array, a 1.x-dimensional array (e.g., a 1.5D array), or a two-dimensional (2D) array. The array of acoustic elements (e.g., one or more rows, one or more columns, and/or one or more orientations) can be uniformly or independently controlled and activated. The transducer array **112** can be configured to obtain one-dimensional, two-dimensional, and/or three-dimensional images of a patient's anatomy. In some embodiments, the transducer array **112** may include a piezoelectric micromachined ultrasound transducer (PMUT), capacitive micromachined ultrasonic transducer (CMUT), single crystal, lead zirconate titanate (PZT), PZT composite, other suitable transducer types, and/or combinations thereof.

[0059] The beamformer **114** is coupled to the transducer array **112**. The beamformer **114** controls the transducer array **112**, for example, for transmission of the ultrasound signals and reception of the ultrasound echo signals. In some embodiments, the beamformer **114** may apply a time-delay to signals sent to individual acoustic transducers within an array in the transducer array **112** such that an acoustic signal is steered in any suitable direction propagating away from the probe **110**. The beamformer **114** may further provide image signals to the processor **116** based on the response of the received ultrasound echo signals. The beamformer **114** may include multiple stages of beamforming. The beamforming can reduce the number of signal lines for coupling to the processor **116**. In some embodiments, the transducer array **112** in combination with the beamformer **114** may be referred to as an ultrasound imaging component. The beamformer **114** may also be a microbeamformer.

[0060] The processor **116** is coupled to the beamformer **114**. The processor **116** may also be described as a processor circuit, which can include other components in communication with the processor **116**, such as a memory, beamformer **114**, communication interface **118**, and/or other suitable components. The processor **116** is configured to process the beamformed image signals. For example, the processor **116** may perform filtering and/or quadrature demodulation to condition the image signals. The processor **116** and/or **134** can be configured to control the array **112** to obtain ultrasound data associated with the object **105**.

[0061] The communication interface **118** is coupled to the processor **116**. The communication interface **118** may include one or more transmitters, one or more receivers, one or more transceivers, and/or circuitry for transmitting and/or receiving communication signals. The communication interface **118** can include hardware components and/or software components implementing a particular communication protocol suitable for transporting signals over the communication link **120** to the host **130**. The communication interface **118** can be referred to as a communication device or a communication interface module.

[0062] The communication link **120** may be any suitable communication link. For example, the communication link **120** may be a wired link, such as a universal serial bus (USB) link or an Ethernet link. Alternatively, the communication link **120** may be a wireless link, such as an ultra-wideband (UWB) link, an Institute of Electrical and Electronics Engineers (IEEE) 802.11 Wi-Fi

link, or a Bluetooth link.

[0063] At the host **130**, the communication interface **138** may receive the image signals. The communication interface **138** may be substantially similar to the communication interface **118**. The host **130** may be any suitable computing and display device, such as a workstation, a personal computer (PC), a laptop, a tablet, or a mobile phone.

[0064] The processor **136** is coupled to the communication interface **138**. The processor **136** may also be described as a processor circuit, which can include other components in communication with the processor **136**, such as the memory **133**, the communication interface **138**, and/or other suitable components. The processor **136** can be configured to generate image data from the image signals received from the probe **110**. The processor **136** can apply advanced signal processing and/or image processing techniques to the image signals. An example of image processing includes conducting a pixel level analysis to evaluate whether there is a change in the color of a pixel, which may correspond to an edge of an object (e.g., the edge of an anatomical feature). In some embodiments, the processor **136** can form a three-dimensional (3D) volume image from the image data. In some embodiments, the processor **136** can perform real-time processing on the image data to provide a streaming video of ultrasound images of the object **105**.

[0065] The memory **133** is coupled to the processor **136**. The memory **133** can be configured to store patient information, measurements, data, or files relating to a patient's medical history, history of procedures performed, anatomical or biological features, characteristics, or medical conditions associated with a patient, computer readable instructions, such as code, software, or other application, as well as any other suitable information or data. The memory **133** may be located within the host **130**. There may also be an additional external memory, or an external memory in replacement of memory **133**. An external memory may be a cloud-based server or an external storage device, located outside of the host **130** and in communication with the host **130** and/or processor **136** of the host via a suitable communication link as disclosed with reference to communication link **120**. Patient information may include measurements, data, files, other forms of medical history, such as but not limited to ultrasound images, ultrasound videos, and/or any imaging information relating to the patient's anatomy. The patient information may include parameters related to an imaging procedure such a probe position and/or orientation.

[0066] The display **131** is coupled to the processor **136**. The display **131** may be a monitor or any suitable display or display device. The display **131** is configured to display the ultrasound images, image videos, and/or any imaging information of the object **105**.

[0067] The system **100** may be used to assist a sonographer or operator in performing an ultrasound scan. The scan may be performed in a point-of-care setting. In some instances, the host **130** is a console or movable cart. In some instances, the host **130** may be a mobile device, such as a tablet, a mobile phone, or portable computer. In yet other examples the host is a server on a cloud and an external display connects to the host in the cloud. During an imaging procedure, the ultrasound system **100** can acquire an ultrasound image of a region of interest of a subject.

[0068] In particular, the ultrasound system **100** may acquire ultrasound images of an artery. More specifically, the ultrasound system **100** may be configured to acquire ultrasound imaging data in accordance with method **300** discussed below including ultrasound data from B-mode ultrasound, Pulsed Wave Doppler mode ultrasound and color or power Doppler mode ultrasound. These three ultrasound modes may be obtained from a single instruction in a “so-called” triplex ultrasound mode, in which three ultrasound modes are used simultaneously to acquire ultrasound data in each of the three modes. In a preferred *modus operandi*, the ultrasound probe will be placed approximately perpendicular, and at an angle of approximately 50 to 70 degrees with respect to the artery, to acquire ultrasound data. It is noted that other angles, between 1 and 89 degrees as also envisaged by the present disclosure.

[0069] FIG. **2** is a schematic diagram of a processor circuit. The processor circuit **200** may be implemented in the probe **110** and/or the host system **130** of FIG. **1**, or any other suitable location,

such as an external computing system separate from an image acquisition device or system. One or more processor circuits can be configured to carry out the operations described herein. The processor circuit **200** can be part of the processor **116** and/or processor **136** or may be separate circuit. In an example, the processor circuit **200** may be in communication with the transducer array **112**, beamformer **114**, communication interface **118**, communication interface **138**, memory **133** and/or the display **131**, as well as any other suitable component or circuit within ultrasound system **100**. As shown, the processor circuit **200** may include a processor **206**, a memory **203**, and a communication module **208**. These elements may be in direct or indirect communication with each other, for example via one or more buses. The processor circuit may further be in communication with an external storage and/or to an external display. The external storage may be used to retrieve or store data, including images, ultrasound data and/or patient information. The external display may be used to display outputs and/or to control the processing circuit, for example by giving instructions.

[0070] A processor **116**, **136**, **206** as envisaged by the present disclosure may include a central processing unit (CPU), a graphical processing unit (GPU), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a controller, a field programmable gate array (FPGA) device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein. A processor **116**, **136**, **206** may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. The processor **260** may also implement various deep learning networks, which may include a hardware or a software implementation. The processor **206** may additionally include a preprocessor in either hardware or software implementation.

[0071] A memory **118**, **138**, **208** as envisaged by the present disclosure may be any suitable storage device, such as a cache memory (e.g., a cache memory of the processor **116**, **136**, **206**), random access memory (RAM), magnetoresistive RAM (MRAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read only memory (EPROM), electrically erasable programmable read only memory (EEPROM), flash memory, solid state memory device, hard disk drives, other forms of volatile and non-volatile memory, or a combination of different types of memory. The memory may be distributed among multiple memory devices and/or located remotely with respect to the processing circuit. In an embodiment, the memory **203** may store instructions **205**. The instructions **205** may include instructions that, when executed by a processor **116**, **136**, **206**, cause the processor **116**, **136**, **206** to perform the operations described herein with reference to the probe **110** and/or the host **130** (FIG. 1).

[0072] Instructions **205** may also be referred to as code. The terms “instructions” and “code” should be interpreted broadly to include any type of computer-readable statement(s). For example, the terms “instructions” and “code” may refer to one or more programs, routines, sub-routines, functions, procedures, etc. “Instructions” and “code” may include a single computer-readable statement or many computer-readable statements. Instructions **205** may be in the form of an executable computer program or script. For example, routines, subroutines and/or functions may be defined in a programming language including but not limited to C, C++, C#, Pascal, BASIC, API calls, HTML, XHTML, XML, ASP scripts, JavaScript, FORTRAN, COBOL, Perl, Java, ADA, .NET, and the like. In particular, according to the present disclosure, instructions as disclosed with reference to method **300** are envisaged herein.

[0073] The communication module **208** can include any electronic circuitry and/or logic circuitry to facilitate direct or indirect communication of data between the processor circuit **200**, the probe **110**, the host **130**, and/or the display **131**. In that regard, the communication module **208** can be an input/output (I/O) device. In some instances, the communication module **208** facilitates direct or indirect communication between various elements of the processor circuit **200**, the probe **110** (FIG. 1) and/or the host **130** (FIG. 1).

[0074] According to an exemplary embodiment, the computer-implemented method contains the steps as disclosed in FIG. 3.

[0075] In step **310**, ultrasound imaging data of an artery is received. The ultrasound data may be received in real time, i.e., during image acquisition, or may be queried upon request from a user or machine from a storage facility, for example, a storage or memory within the image acquisition device, a cloud, or a physical hard drive external to the image acquisition device. According to the invention, the ultrasound data comprises any combination including: [0076] B-mode ultrasound data, [0077] pulsed wave Doppler ultrasound data, and [0078] color or power Doppler ultrasound data.

It is understood by a person skilled in the art that other forms of ultrasound data, in addition and/or in replacement of the above ultrasound data may also be received.

[0079] According to an embodiment of the invention, the ultrasound data may be received from an ultrasound device or system as provided in FIG. 1 comprising an array of acoustic elements **112** configured to acquire [0080] B-mode ultrasound data, [0081] pulsed wave Doppler ultrasound data, and [0082] color or power Doppler ultrasound data.

[0083] The acquisition mode may be pre-programmed, e.g., as a triplex ultrasound imaging mode, in such a manner that the [0084] B-mode ultrasound data, [0085] pulsed wave Doppler ultrasound data, and [0086] color or power Doppler ultrasound data

are obtained from a sequential or alternating ultrasound shots. This pre-programmed control may be executed upon a user request to use said triplex mode, or upon a machine command in case a velocity profile is selected to be determined.

[0087] According to an embodiment, the ultrasound device comprising an acoustic array configured for triplex mode as described above, may be a hemodynamic 1D ultrasound patch, e.g., an ultrasound patch that may be attached on a subject's skin for continued monitoring of a feature within the subject (e.g., heart, lungs, etc.). In such a case, the array of acoustic elements may acquire data continuously in triplex mode (e.g., by acquiring ultrasound data simultaneously in each mode or by alternating the different modes within the triplex mode). In other words, acquire data of sufficient time length in for example, first B-mode, then pulsed wave Doppler mode and last in color or power Doppler mode, and then repeat acquisition. It should be understood that any order of these modes is envisaged by the invention.

[0088] In step **320**, the ultrasound data is processed in order to determine a first set of properties or measurements from the artery. In particular, in step **320** [0089] a dimension of an artery, [0090] a velocity in an artery and [0091] a measure of flow profile asymmetry are determined from the ultrasound data.

[0092] An artery dimension (or lumen dimension) refers to any dimension relating to a cross-section of the artery. An artery dimension (or lumen dimension) may thus, for example, be the artery diameter or radius. The artery dimension may however also be for example the surface area or perimeter of a transversal cross-section of the artery. The artery or lumen dimension may be obtained from for example B-mode ultrasound data.

[0093] FIG. 4 displays an ultrasound image **410** generated from B-mode ultrasound data of a cross-section of an artery according to an embodiment of the invention. In particular, FIG. 4 displays a cross-section of an artery at approximately 20 to 40 degrees with respect to the length of the artery and approximately 50 to 70 degrees with respect to a transversal cross-section, such that the diameter or radius may be obtained from the minor axes **412** of the illustrative oval **414** drawn merely for illustrative purposes. It should be clear to a person skilled in the art, that other ultrasound data may also be suitable to determine a dimension of the artery, such as for example color or power Doppler ultrasound data shown in FIG. 8. It should also be clear to a person skilled in the art, that artery dimension and lumen dimension in the context of this invention may be used interchangeably.

[0094] According to an embodiment of the invention, a velocity in the artery refers to a local

velocity as measured in an area smaller than the area of a transversal artery cross-section. A velocity in the artery is thus a local velocity at any given location or area within the lumen of the artery. For example, a velocity in an artery may be the velocity in the center of the artery. According to another example, a velocity may be the velocity at a pre-defined location within the artery, such as for example the location of the maximum velocity within the artery, or a location near an edge of the artery. A velocity in the artery may be determined from any ultrasound mode, such as color Doppler, power Doppler or pulsed wave Doppler ultrasound. In an embodiment, a velocity is determined from pulsed wave Doppler ultrasound data. The pulsed wave Doppler ultrasound mode, for example, allows to measure a local velocity at a higher accuracy than in color or power Doppler ultrasound mode. Due to the high frame rate, pulsed wave Doppler ultrasound mode, may thus also be advantageous to resolve important features of the velocity waveform, e.g., the systolic peak velocities, in an artery.

[0095] FIG. 5 shows visualized ultrasound data **500** obtained from a pulsed wave Doppler ultrasound mode. The ultrasound data includes an ultrasound image **510** and a timeline of a velocity **520**, wherein the velocity is determined within a relatively small sampling region **530**. The timeline displays an evolution of the velocity within the sampling area **530** over time, resolving, due to the high frame rate of pulsed wave Doppler, systolic peaks **522**. A velocity at a given location within the artery or lumen, may thus be determined as an instantaneous velocity, e.g., a single point from the timeline **520**, or as an average velocity, averaging over the velocity in the timeline **520**. Additionally, the sampling region **530**, may be as shown in FIG. 5, at the center of the artery, or at any point within the cross-section, such as for example the maximum velocity location determined from color Doppler-mode ultrasound data.

[0096] According to an embodiment of the invention, a measure of asymmetry of the flow refers to any parameter which indicates a level of asymmetry or skewedness in a flow profile with respect to a symmetric flow profile. For example, a measure of asymmetry may be a deviation of a flow profile from a symmetric parabolic flow profile. In another example, a measure of asymmetry may be a deviation of the velocity maximum from the center axes of the artery or centroid of an artery cross-section. For example, in curved arteries, so-called Dean vortices may arise and skew the typically symmetric parabolic flow profile observed in straight arteries, in such a way that the maximum velocity within the artery may no longer lie at the central axes of the artery, but at a certain distance, shifted away from the central axes of the artery. Such deviations of the flow are shown in FIGS. **6A** and **6B**.

[0097] FIG. **6A** displays a 2D velocity distribution in a curved vessel as a contour plot. Each line in the contour plot represents a particular velocity. For example, starting at a zero velocity for the most outer line, each next line towards the center of the Fig represents an increase velocity.

[0098] In a typically straight artery, the velocity profile may be entirely symmetric across both shown axes **680** and **690** (these axes may be at any orientation, as long as the two axes are perpendicular to each other). In curved arteries however, due to the Dean vortices, the velocity distribution may shift to be asymmetric across at least one axes **690**. The maximum velocity, being in the center of the area delimited by the most inner contour line **640**, thus shifts from the centroid **670** by a distance **632** to a new location **630**. The distance that the velocity maximum shifts is determined by the radius of curvature $R_{sub.c}$ **734** of the artery **700** as shown in FIG. 7.

[0099] FIG. **6B** displays a velocity profile along the asymmetric axis **680** for various radii of curvature **734**. For example, an artery with a zero radius of curvature would display a fully symmetric flow profile **631**. As the radius of curvature increases the velocity maximum shifts by a distance **632**, wherein the distance **632** increases as the radius of curvature increases.

[0100] FIG. **8** displays an ultrasound image **810** from color or power Doppler ultrasound data with a heat map **850** of the velocity in a cross-section of the artery. While the heat map may be qualitative, e.g., without any indication of absolute velocity values, it may provide information with regards to the velocity maximum **630** due to a change in color, color intensity and/or

brightness at the maximum. The shift **632** of the velocity maximum **630** with respect to the centroid **670** of the artery cross-section may then be determined by, for example, calculating a distance **632** between the velocity maximum location **630** and the centroid **670** of the artery cross-section.

[0101] According to an embodiment of the invention, the velocity maximum shift **632** may be associated to a non-dimensional parameter such as a local Dean number De . In particular, it has been shown for at least self-similar flows in, for example, Cieřlicki et al. (2012) “Can the Dean number alone characterize flow similarity in differently bent tubes?”, Verkaik et al. (2009) “Estimation of volume flow in curved tubes based on analytical and computational analysis of axial velocity profiles”, and Petrakis et al. (2009) “Steady Flow in a Curved Pipe with Circular Cross-Section. Comparison of Numerical and Experimental Results”, that the shift dx **632** of the velocity maximum location is dependent on the Dean number De according to FIG. **9**. In other words, the local Dean number may be extracted from for example FIG. **9** by merely having determined the shift **632** of the maximum velocity, without requiring a value for the velocity. For example, the maximum velocity shift may be determined to be 40% of the artery diameter, such that the local Dean number corresponds to approximately 200.

[0102] It is noted that the present invention also envisages approaches in which the Dean number as obtained according to the velocity shift and FIG. **9** may only be a first estimation and may be re-calculated at a later stage. The Dean number may for example be determined iteratively to converge to a value by combining blocks of the method **300** in a different order, and/or adding and/or removing blocks of the method **300**.

[0103] With reference again to FIG. **3**, in step **330**, a flow profile within the artery is determined. For example, the color or power Doppler ultrasound data, may be integrated pixel by pixel over the heat map **850** of FIG. **8**. In another example, a flow profile is determined based on the first set of properties as determined in step **320** in combination with an analytic expression. For example, an artery dimension, a velocity in an artery and a measure of asymmetry may be combined to determine a flow profile. In the following various suitable methods will be presented to determine a flow profile. It should however be clear that methods deviating from the below, but within the same spirit, are also envisaged by the present invention.

[0104] According to an embodiment based on Soedberg (1987) “Viscous flow in curved tubes—I. velocity profiles”, a non-dimensional or normalized flow profile $u(r)$, passing through the maximum velocity, may be determined from an artery dimension and a measure of asymmetry following

$$[00001] \quad u(r) = w_0(r) - w_1(r) + \dots \text{Math.} \quad (1) \quad \text{where two first terms are given by } w_0(r) = \frac{(1 - r^{F-1})F}{(F-1)}, \text{ and } w_1(r) = 1 - 0.3r - 0.7r^{\sqrt{q} - 0.95}.$$

[0105] Here $w_{\text{sub}.0}$ and $w_{\text{sub}.1}$ represent the zeroth and first tangential harmonic contributions to the flow. Higher harmonic contributions may also be considered, such as for example the second, third and so on harmonic contributions. In addition or in replacement, also different expressions of for the harmonic contributions may be used such as

$$[00002] \quad w(r) = 2 \frac{(1 - r^{F-1})F}{(F-1)},$$

[0106] Further, r represents a radial coordinate normalized by the total artery radius R (i.e., $r = r'/R$), which may be derived as described above. It should be noted, a velocity profile along any radial direction of a cross-section may be obtained. For example, one velocity profile $u(r)$ may be along the axis **680** in FIG. **6A**, and another velocity profile $u(r)$ may be along the axis **690** shown in FIG. **6A**.

[0107] Further, F represents a form factor which may be any one of:

$$[00003] \quad F = 0.5 \text{ Math. } q^{0.6} + 1.5, F = 0.5(1 + q^{0.5}) + 1.5, F = 0.5(1 + q)^{0.6} + 1.5, \text{ with } q = De / 16,$$

where De is the local Dean number, representative of a measure of asymmetry, and which may be derived from the shift of the maximum velocity as discussed above under step **320** of the method

300.

[0108] According to an embodiment based on Tijssen (1979) “Axial dispersion in helically coiled open columns for chromatography”, the flow profile may be estimated according to

$$[00004] \quad u(r) = q^2 (1 - r)^2 \exp\{(1 - r)(\ln(8 / 9) - 0.5\ln[(q / 1.6)^{3.4} + 1])\} \quad (2) \quad \text{or}$$

$$u(r) = \frac{F^2}{F-1} (1 - r^F) \exp[(1 - r^2)(1 - a\ln F^b)] \quad (3)$$

where in addition to the variable definitions above a is typically between 0.2 and 0.3, preferably 0.25 and b is between 3.5 and 4, preferably 3.75. It should be understood that a and b are fitting constants, which may thus deviate from the mentioned values, may be removed or replaced by different fitting constants.

[0109] Following the above expressions for $u(r)$, the obtained velocity profile may be accurate in shape, but not in absolute values, given that it concerns a normalized velocity profile. Therefore, the normalized velocity profile may be shifted or scaled by a velocity in an artery, for example, by a velocity, derived from a pulsed wave Doppler measurement. In such a case, the normalized velocity profile $u(r)$, which may be any velocity profile as shown in FIG. 6B, may be shifted from non-dimensional normalized units to having units of for example distance over time (i.e., m/s, cm/s, feet/s, etc.).

[0110] The inventors have further appreciated that the accuracy of the velocity distribution obtained following Eq. (1) may be within 10% of the real velocity distribution when $16 < De < 240$, and that the accuracy of the velocity distribution obtained following Eqs. (2) or (3) may be within 10% of the real velocity distribution when $De > 100$.

[0111] According to an embodiment, Eq. (1) and either of Eqs. (2) or (3) may thus be combined, such that for lower De numbers Eq. (1) is used and for higher De numbers either of Eqs. (2) or (3) may be used in order to obtain an accuracy of the velocity profile within 10% for a greater De number range. The relevant De range within an artery may for example be $1 < De < 2000$.

[0112] According to an embodiment, the flow profile (this may be the non-dimensional normalized flow profile or the scaled and dimensional flow profile) may further be used to determine a 3D flow profile as proposed in Verkaik et al. (2009). The cross section (e.g., as shown in FIG. 6A) is divided into two semicircles along the diameter perpendicular to that on which the flow profile is known. The flow rate $Q(r, \theta)$ is then estimated by assuming the axial flow to be axisymmetric in each semicircle as shown in FIG. 10, giving the expression

$$[00005] Q(r, \theta) = \int_0^R u^+(r) r dr + \int_0^R u^-(r) r dr.$$

This expression may be solved according to integration by parts.

[0113] In an optional step **340**, a radius of curvature $R_{\text{sub.c}}$ **734** may be determined from the flow profile $u(r)$.

[0114] i) From the flow profile determined in step **130**, a Reynolds number Re may be determined following

$$[00006] Re = \frac{U_{2R}}{\rho}.$$

Here ρ is the blood density, U is the average velocity in the artery at the imaged cross-section, D and μ is the dynamic viscosity. The average velocity may be obtained as: [0115] half the maximum velocity for laminar flows, [0116] the center velocity for turbulent flows, or [0117] as the average velocity of the entire cross-section, which may be obtained through mathematical averaging.

[0118] ii) The radius of curvature may be obtained relating the Reynolds number to the Dean number according to

$$[00007] De = Re \sqrt{\frac{2R}{R_c}}.$$

[0119] It is understood by a person skilled in the art that while method **300** was presented in a specific order, the order of the steps may vary, additional steps may be added in between, and/or steps may be removed.

[0120] It is furthermore understood that the individual steps in the method **300** may be performed

in real time, i.e., during an examination procedure, or thereafter. For example, the ultrasound data may be obtained in real time and analyzed according to method **300** during an image acquisition procedure, or the imaging data may be first obtained during an acquisition step and only be analyzed upon request of an operator and/or expert. The method **100** may thus be implemented within an image acquisition system (FIG. **1**) or a hemodynamic patch or outside of it in a separate computing unit (FIG. **2**). In another embodiment the method **100** may also be carried out in a cloud.

[0121] It should be noted that the above-mentioned embodiments illustrate rather than limit the invention, and that those skilled in the art will be able to design many alternative embodiments without departing from the scope of the appended claims. In the claims, any reference signs placed between parentheses shall not be construed as limiting the claim. The word “comprising” does not exclude the presence of elements or steps other than those listed in a claim. The word “a” or “an” preceding an element does not exclude the presence of a plurality of such elements. The invention may be implemented by means of hardware comprising several distinct elements, and/or by means of a suitably programmed processor. In the device claim enumerating several means, several of these means may be embodied by one and the same item of hardware. Measures recited in mutually different dependent claims may advantageously be used in combination.

Claims

1. A computer-implemented method for determining a flow profile in an artery based on ultrasound imaging data, the method comprising: receiving ultrasound imaging data of an artery in triplex mode, wherein the triplex mode comprises: B-mode ultrasound data, pulsed wave Doppler ultrasound data, and color or power Doppler ultrasound data; determining, from the ultrasound imaging data, an artery dimension; a velocity in the artery cross-section, and a measure of asymmetry of the flow profile; and determining a flow profile from the artery dimension, the velocity in the artery cross-section, and the measure of asymmetry of the flow profile.
2. The method of claim 1 further comprising: determining a local artery curvature based on the artery dimension, the flow profile, and the measure of asymmetry of the flow profile.
3. The method of claim 1, wherein the artery dimension is determined based on the B-mode ultrasound data.
4. The method of claim 1, wherein the velocity in the artery cross-section is determined based on the pulsed wave Doppler ultrasound data.
5. The method of any of claim 1, wherein the velocity in the artery cross-section is determined in a center of the artery cross-section.
6. The method of any of claim 1, wherein the measure of asymmetry of the flow profile is determined based on the color or power Doppler ultrasound data.
7. The method of claim 1, wherein the measure of asymmetry comprises: determining a location of the maximum flow velocity in an artery cross-section; and based on said location, determining a parameter representative of a deviation of said location with respect to the center of the artery cross-section.
8. The method of claim 7, wherein the parameter representative of the deviation of the location of the maximum flow velocity in an artery cross-section is the Dean number.
9. A computer program product comprising instructions for enabling a processor to carry out the method of claim 1.
10. A system for determining a flow profile from ultrasound imaging data, the system comprising a processor configured to carry out the method of claim 1.
11. The system of claim 10, further comprising: an array of acoustic elements in communication with the processor, and wherein the processor is configured to control the array of acoustic

elements to send and receive acoustic signals to generate ultrasound imaging data.

12. The system of claim 10, wherein the system is a hemodynamic monitoring patch.
